Project data dictionary

NYPD Complaint Data, fixed slice: complaints reported from 1 January 2020 to 30 June 2026

1. The file

The project data come in one file, `nypd_2020_2026.rds`, available on Moodle. It holds every complaint recorded by the New York City Police Department (NYPD) and reported between 1 January 2020 and 30 June 2026: 3 386 974 rows and 39 columns. One row is one complaint, i.e. one offence reported to the police, completed or attempted. The slice was extracted on 14 October 2026 and will not change during the semester, so every group works on exactly the same data.

Source: NYC Open Data, "NYPD Complaint Data Historic" (dataset qgea-i56i) for complaints reported from 2020 to 2025, and "NYPD Complaint Data Current (Year To Date)" (dataset 5uac-w243) for 2026. Compared with the published data, column names were converted to lower case, the coordinate columns in the New York State Plane system were dropped (latitude and longitude are kept), and seven columns were added to save you some parsing (marked "derived" in the tables below). The values themselves are as published, including their spelling and their gaps.

Reading the file

```
library(dplyr)
nypd <- readRDS("nypd_2020_2026.rds")
nypd <- as.data.frame(nypd)
str(nypd)
```

An `.rds` file stores one R object exactly as it was in memory, with its column types: dates are dates, `hour` is an integer, `wday` is an ordered factor from Monday to Sunday. There is nothing to parse and no separator or encoding to guess. Any R installation reads it with `readRDS()`, with no extra package. If you ever need the data in another tool, `readr::write_csv(nypd, "nypd.csv")` exports them as text.

The third line converts the tibble that `readRDS()` returns into a plain `data.frame`. The reason is practical: on some Windows 11 laptops, printing a tibble fails with the message "An Application Control policy has blocked this file", because a DLL used by the tibble package is blocked by the system. A `data.frame` is printed by base R and avoids the problem, and dplyr and ggplot2 work in the same way with both. For the same reason this guide uses `str(nypd)` instead of `glimpse(nypd)`: both print one line per column with its type and first values. Typing `nypd` alone on the console prints hundreds of rows, so use `head(nypd)` to look at the first ones.

Keep the data file out of the weekly ZIP, since the project protocol asks for the project folder without it. Summarise before you plot: 3.4 million points on one scatterplot are slow to draw and say little, while counts by hour, by premises type, by month or by precinct are small tables that ggplot2 draws instantly.

After reading the file, `nrow(nypd)` should return **3 386 974**. If it does not, you have a different file.

2. The columns

The 39 columns are grouped below by the question they answer: when, where, what, who. The type is the R class in the file. Example values are a sample, never the full list; use `count(nypd, column, sort = TRUE)` to see every value and its frequency. Columns marked "derived" were computed from the published ones and are the ones to use for dates and times.

When: dates and times

Column	Type	Meaning and notes
`cmplnt_num`	text	Identifier of the complaint. Unique in this file (one row per complaint). Keep it as text: it is a label, not a number.
`rpt_dt`	text	Date the complaint was reported to the police, in the export format of NYC Open Data (for example `2020-03-14T00:00:00.000`). Use `reported_on` instead.
`cmplnt_fr_dt`	text	Date the offence occurred or started, in the same export format. Use `occurred_on` or `occurred_at` instead.
`cmplnt_fr_tm`	text	Time the offence occurred or started, `HH:MM:SS`. Use `hour` or `occurred_at` instead.
`cmplnt_to_dt`, `cmplnt_to_tm`	text	End date and time, filled when the offence spans a period (a burglary discovered after a weekend away, for instance). Empty for most rows.
`reported_on`	Date (derived)	`rpt_dt` as an R date. This column defines the slice: all values lie between 2020-01-01 and 2026-06-30.
`occurred_on`	Date (derived)	`cmplnt_fr_dt` as an R date. Can be earlier than 2020 (see `old_occurrence`).
`occurred_at`	date-time, POSIXct (derived)	Date and time of occurrence combined. NA when the time is missing.
`hour`	integer, 0 to 23 (derived)	Hour of occurrence. NA when the time is missing. Before trusting a peak, count the rows with the time exactly `00:00:00` or `12:00:00`: an unknown time is sometimes recorded as a round one.
`wday`	ordered factor, Mon to Sun (derived)	Weekday of occurrence, week starting on Monday. Keep the factor as it is: it is what puts the days in the right order in your plots.
`report_year`	number (derived)	Year of `reported_on`, 2020 to 2026.
`old_occurrence`	logical (derived)	TRUE when `occurred_on` is before 1 January 2020 (16 630 rows). Mixes genuine late reports with impossible dates (years such as 1015 or 1900). Decide what to do with these rows before any analysis of occurrence dates.

Where

Column	Type	Meaning and notes
`boro_nm`	text	Borough: BRONX, BROOKLYN, MANHATTAN, QUEENS, STATEN ISLAND. NA in 5 835 rows.
`addr_pct_cd`	text	Number of the police precinct (1 to 123, with gaps). A code, not a quantity: do not average it.
`patrol_boro`	text	Patrol borough, the eight NYPD commands (for example PATROL BORO BKLYN NORTH, PATROL BORO MAN SOUTH, PATROL BORO QUEENS SOUTH).

Column	Type	Meaning and notes
latitude, longitude	number	Coordinates of the occurrence in decimal degrees (WGS 84). NA in 28 rows. Locations are geocoded to a nearby reference point (an intersection or the middle of the block) rather than to the exact address.
prem_typ_desc	text	Type of premises where the offence occurred: STREET, RESIDENCE - APT. HOUSE, RESIDENCE-HOUSE, RESIDENCE - PUBLIC HOUSING, COMMERCIAL BUILDING, CHAIN STORE, GROCERY/BODEGA, TRANSIT - NYC SUBWAY, PARK/PLAYGROUND, BAR/NIGHT CLUB, RESTAURANT/DINER and about sixty more. The central column for questions about the kind of place.
loc_of_occur_desc	text	Position relative to the premises: INSIDE, OUTSIDE, FRONT OF, REAR OF, OPPOSITE OF. Empty for many rows.
parks_nm	text	Name of the park, when the offence occurred in one. Empty otherwise.
hadevelopt	text	Name of the NYCHA public housing development, when the offence occurred in one.
housing_psa	text	Police Service Area code, for public housing.
station_name	text	Subway station, for offences in the transit system.
transit_district	text	Transit police district number.

What

Column	Type	Meaning and notes
law_cat_cd	text	Level of offence: FELONY, MISDEMEANOR or VIOLATION, from the most to the least serious.
ky_cd	text	Three-digit offence classification code. Each code has one description in ofns_desc.
ofns_desc	text	Description of the offence class: PETIT LARCENY, HARRASSMENT 2, ASSAULT 3 & RELATED OFFENSES, GRAND LARCENY, CRIMINAL MISCHIEF & RELATED OF, FELONY ASSAULT, ROBBERY, BURGLARY, DANGEROUS DRUGS and others. Spelling and abbreviations are as published.
pd_cd	text	Internal classification code, finer than ky_cd.
pd_desc	text	Description of the internal classification (several hundred values, for example LARCENY,PETIT FROM STORE-SHOPL). More specific than ofns_desc.
crm_atpt_cptd_cd	text	Whether the offence was COMPLETED or ATTEMPTED.
juris_desc	text	Agency with jurisdiction over the offence: N.Y. POLICE DEPT, N.Y. TRANSIT POLICE, N.Y. HOUSING POLICE, PORT AUTHORITY and others.
jurisdiction_code	text	Code for juris_desc: 0 police, 1 transit, 2 housing, 3 and above for agencies outside the NYPD.

Who: suspect and victim

Column	Type	Meaning and notes
`susp_age_group`, `vic_age_group`	text	Age group of the suspect and of the victim: <18, 18-24, 25-44, 45-64, 65+, UNKNOWN. Any other value (a negative number, a year) is a recording error.
`susp_race`, `vic_race`	text	Race as recorded by the police officer, for example WHITE HISPANIC, BLACK, WHITE, ASIAN / PACIFIC ISLANDER, BLACK HISPANIC, AMERICAN INDIAN/ALASKAN NATIVE, UNKNOWN. These are classifications made by the police, not self-identification, and should be presented as such.
`susp_sex`, `vic_sex`	text	M, F or U (unknown). In `vic_sex`, D means the victim is a business or organisation and E the People of the State of New York, so neither is a person.

3. Before you analyse

- The slice is defined by the report date. An offence that occurred in December 2019 and was reported in January 2020 is in the file; one that occurred in June 2026 and was reported in July is not. For questions about when offences happen, use the occurrence columns. For questions about police workload, use the report date.
- Missing values appear as NA, but some categorical columns also carry the literal text "(null)" or the value UNKNOWN. Check each column you use with `count()` and decide how to treat these values before counting.
- The file records reported offences. It says nothing about arrests, about outcomes, or about offences that were never reported, and one row is one complaint, not one victim.
- Hours and coordinates are recorded by people. A spike at a round hour or a pile of points on one coordinate deserves a look before you draw conclusions from it.

4. First commands

A few lines to get a feel for the data before you decide on your questions:

```
count(nypd, report_year)
count(nypd, boro_nm, sort = TRUE)
count(nypd, law_cat_cd)
count(nypd, ofns_desc, sort = TRUE) |> slice_head(n = 20)
count(nypd, prem_typ_desc, sort = TRUE) |> slice_head(n = 20)
count(nypd, hour)
nypd |> filter(!old_occurrence) |> count(wday)
```

5. Source and further documentation

NYC Open Data, NYPD Complaint Data Historic (data.cityofnewyork.us, dataset qgea-i56i) and NYPD Complaint Data Current (Year To Date) (dataset 5uac-w243). The data are published under the NYC Open Data terms of use and may be used freely with attribution. Each dataset page links to the official column notes ("NYPD Complaint Incident Level Data Footnotes"), which remain the authority on the meaning of every code.

> Cite the source in every figure caption: "Data: NYPD Complaint Data, NYC Open Data (complaints reported from January 2020 to June 2026)".